

Ayushman Lohani

GenAI Engineer | Agentic RAG & LLM Systems | PyTorch, LangChain, FastAPI

+91 6393095246 | aayushmanlohani@gmail.com | [linkedin.com](https://www.linkedin.com) | github.com

EDUCATION

University of Lucknow

Bachelor of Technology in Computer Science (Artificial Intelligence)

Lucknow, UP

Oct. 2023 – Sep 2027

EXPERIENCE

AI & Machine Learning Intern

July 2025 – Sep. 2025

National Institute of Electronics & Information Technology (NIELIT)

Online

- Built a sequence-to-sequence Transformer from scratch in PyTorch, implementing Multi-Head Attention and positional encoding to replicate core generative AI architecture.
- Optimized training across 175,000+ data points using dynamic learning rate schedulers, reducing validation loss to 0.84 and cutting convergence time by 25%.
- Refactored experimental Jupyter notebooks into modular, production-ready Python scripts, enabling real-time model inference and clean deployment handoff.

Machine Learning & Data Science Intern

Jul. 2025 – Aug. 2025

Info Bharat Interns

Online

- Engineered a time-series forecasting pipeline using XGBoost across 17,000+ temporal records, achieving an R^2 score of 0.96 in demand trend prediction.
- Standardized end-to-end data preprocessing workflows including feature scaling and temporal alignment, eliminating data inconsistencies and improving model reliability.

PROJECTS

Unitwise — Agentic RAG Study Assistant

[\[GitHub\]](#) [\[Live\]](#)

Python, LangChain, FastAPI, Vector DB, Groq API

- Engineered a FastAPI backend supporting 10 simultaneous users via API rate limiting and request load balancing, ensuring stable performance under concurrent usage.
- Built an agentic Retrieval-Augmented Generation (RAG) pipeline using LangChain and Vector DB to answer curriculum-based queries, serving 50+ registered users with 1,000+ questions answered collectively.
- Engineered prompt templates and context-window management strategies to optimize LLM response quality while controlling API costs at scale.

RBI Sentinel — Market Volatility Forecaster

[\[GitHub\]](#) [\[Live\]](#)

Python, FinBERT, Scikit-Learn, Pandas, Statsmodels

- Engineered an NLP pipeline using FinBERT to classify hawkish/dovish sentiment across 140+ RBI monetary policy documents, converting central bank language into quantifiable econometric signals.
- Trained a Random Forest classifier to predict India VIX volatility shocks from extracted sentiment features, achieving an R^2 score of 0.91 on held-out test data.
- Aligned time-series data by forward-filling weekend policy releases to Monday market open, validating the information content hypothesis between RBI communications and market movement.

TECHNICAL SKILLS

Languages: Python, SQL, C++

ML/AI Frameworks: PyTorch, Scikit-Learn, LangChain, LangGraph, HuggingFace, XGBoost, FinBERT

Concepts: Large Language Models, RAG, NLP, Agentic AI, Embeddings, Vector Databases, Prompt Engineering

Libraries, Tools: Pandas, NumPy, FastAPI, Streamlit, Git, Jupyter

CERTIFICATIONS

Artificial Intelligence Fundamentals — **IBM**

Certified Data Science Professional — **Oracle**